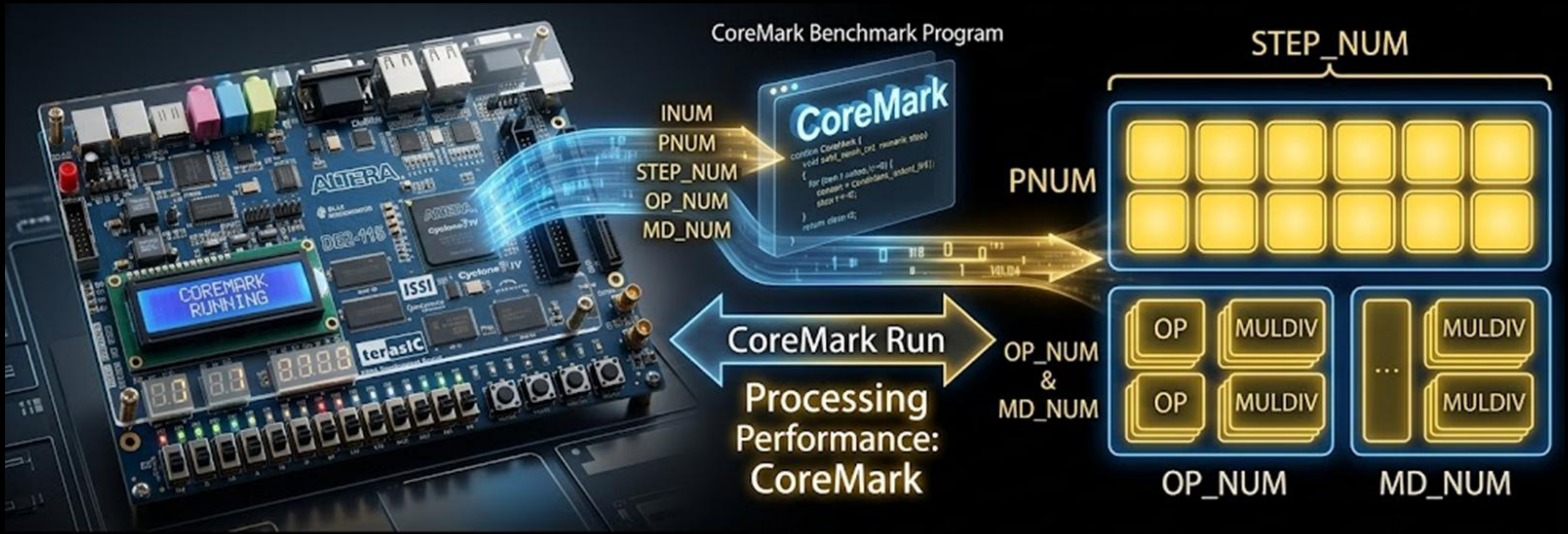
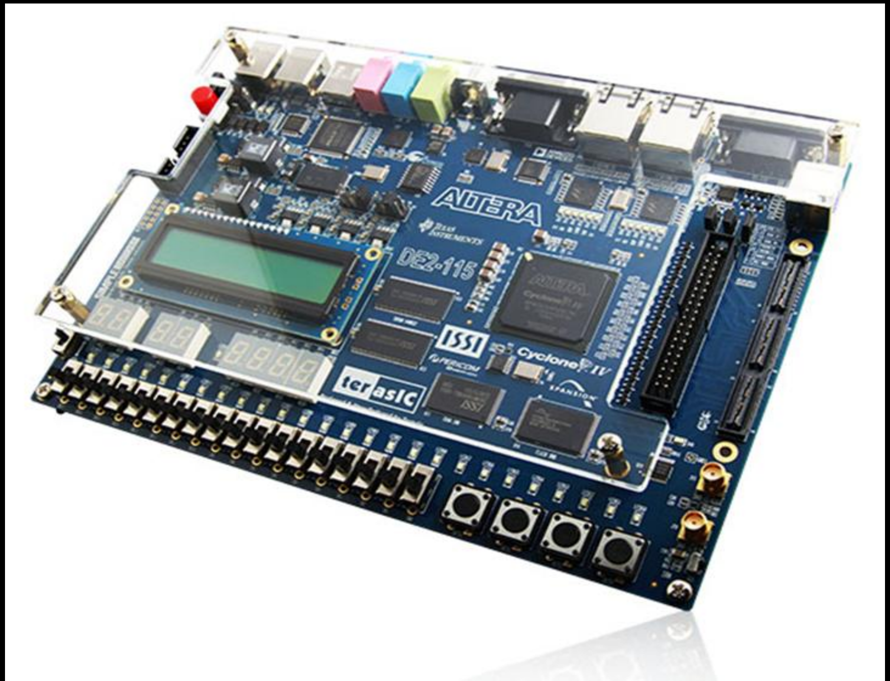


Test Processing Performance: CoreMark via parameterized RV32 cores on a physical FPGA platform. Adjusting core parameters dynamically alters benchmark scores.

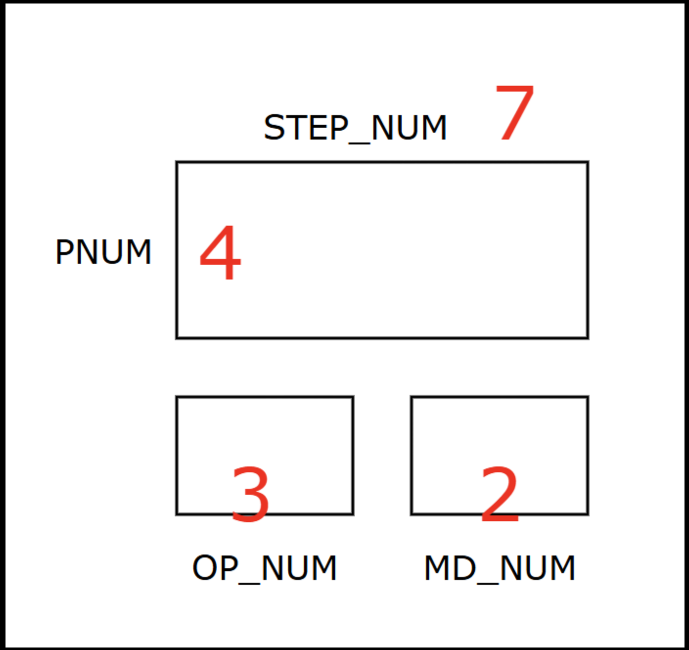
ISM (Instruction Scheduling Matrix): PNUM × STEP_NUM

Execution Units Array: OP_NUM / MD_NUM

Performance Profile: Fully scaled by these 4 core parameters



+



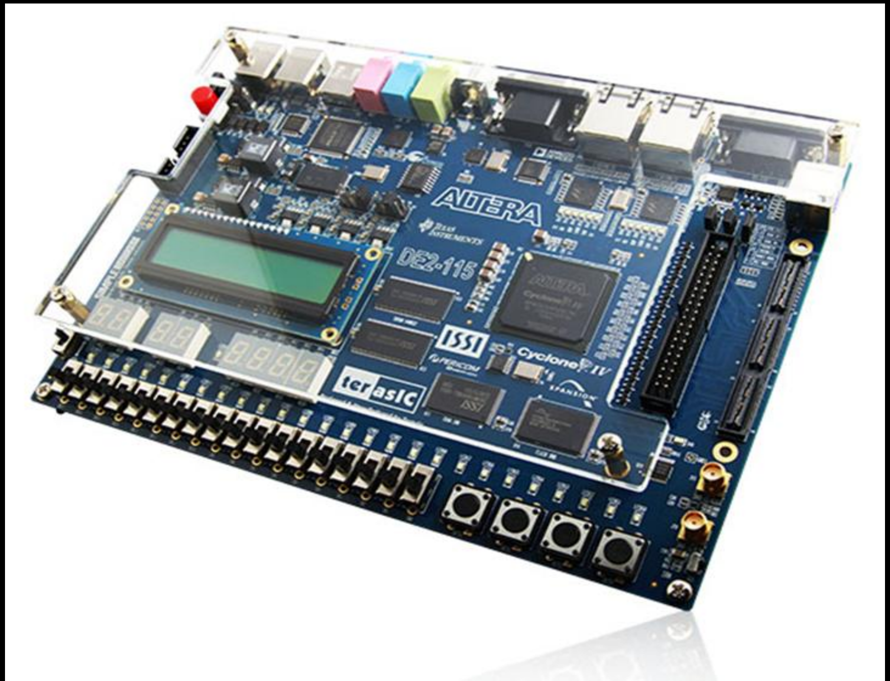
FPGA_25_4_7

FPGA Platform: DE2-115

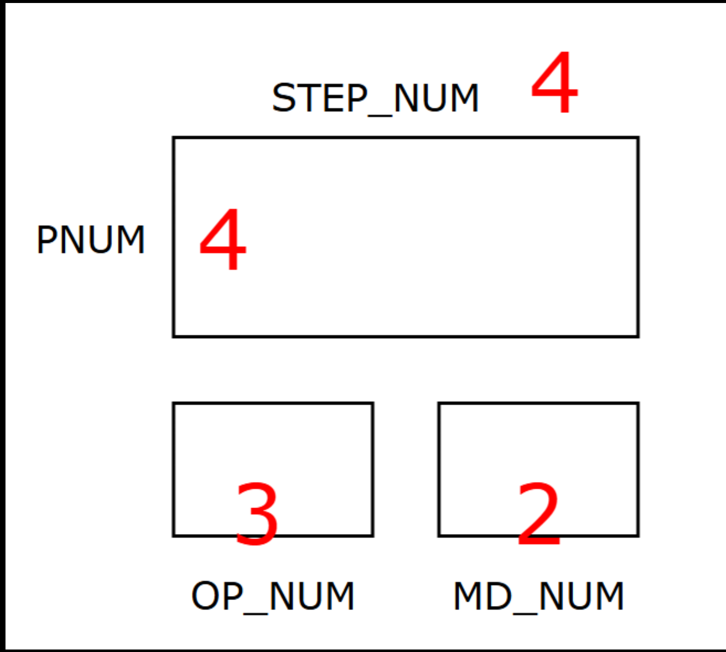
Resource utility: 66%

Clock frequency: 25 MHz

CoreMark/MHz: 7.5



+



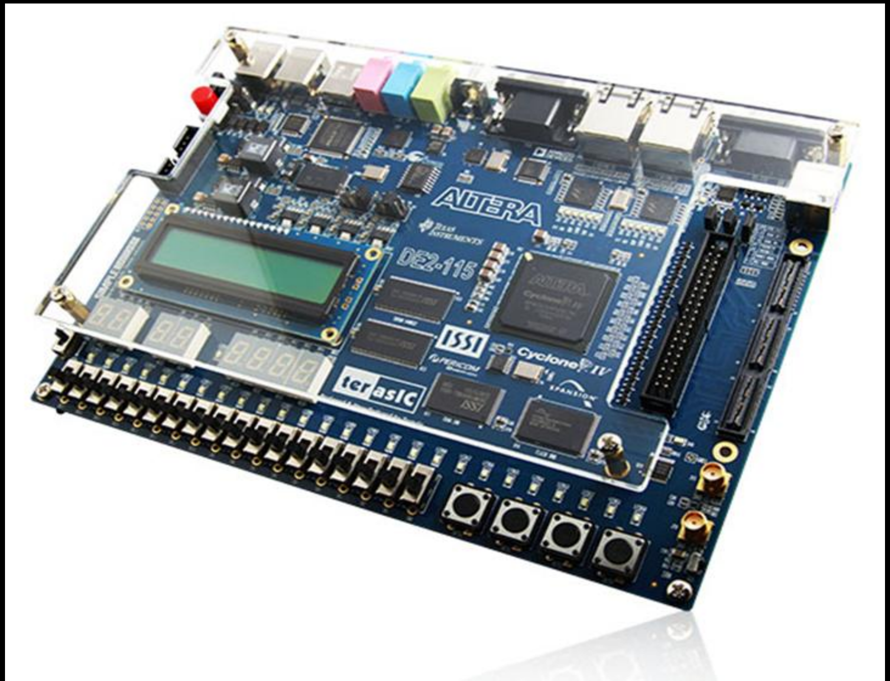
FPGA_33_4_4

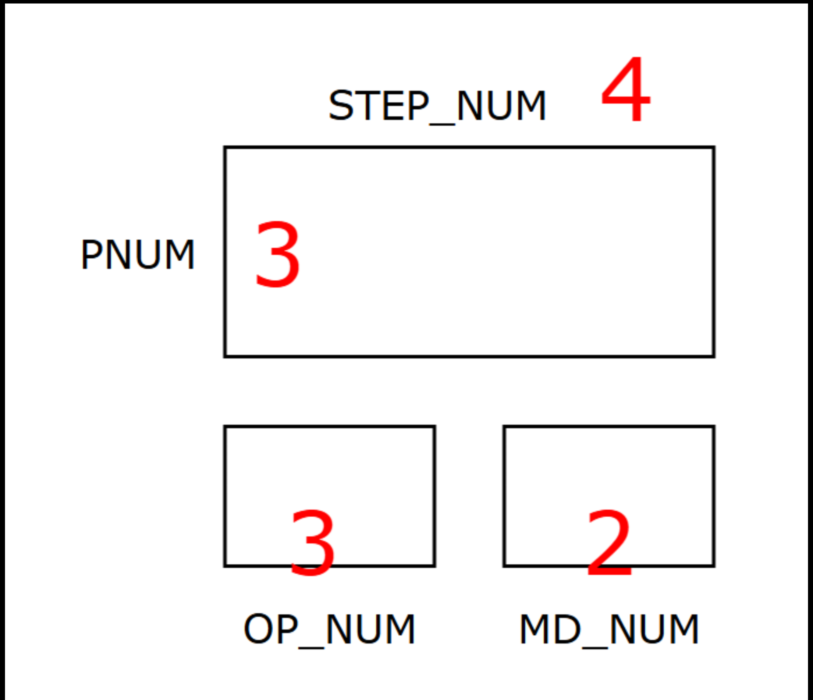
FPGA Platform: DE2-115

Resource utility: 43%

Clock frequency: 33 MHz

CoreMark/MHz: 7.28





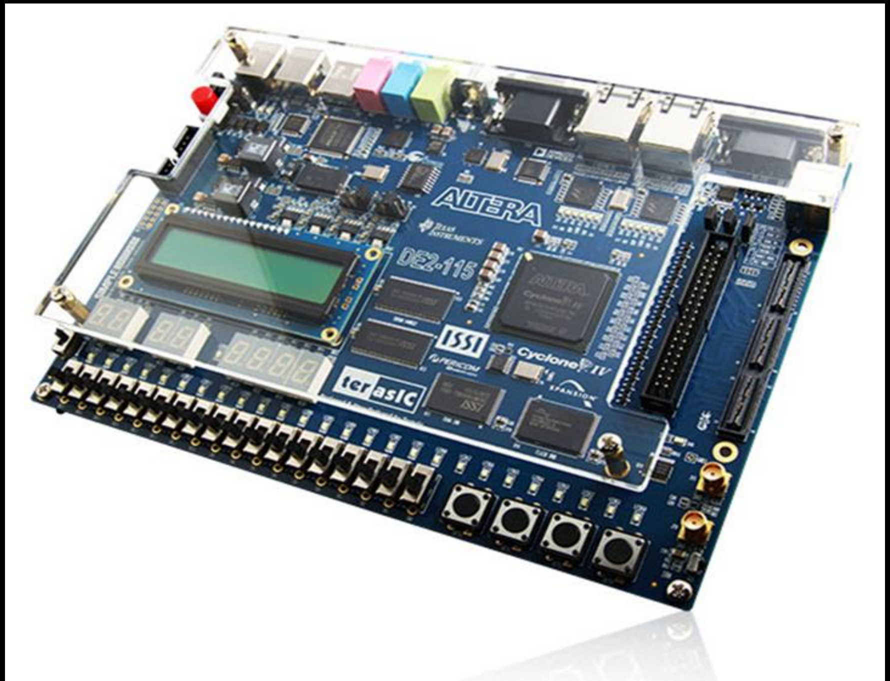
FPGA_40_3_4

FPGA Platform: DE2-115

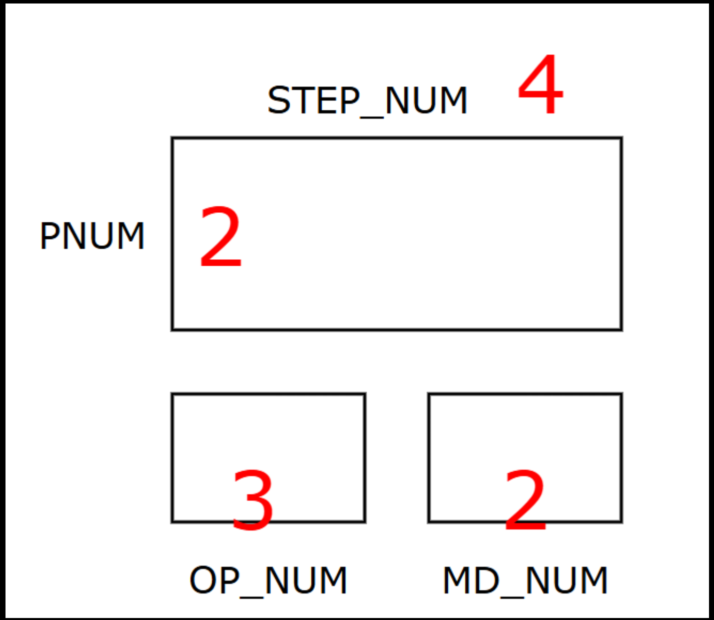
Resource utility: 34%

Clock frequency: 40 MHz

CoreMark/MHz: 6.98



+



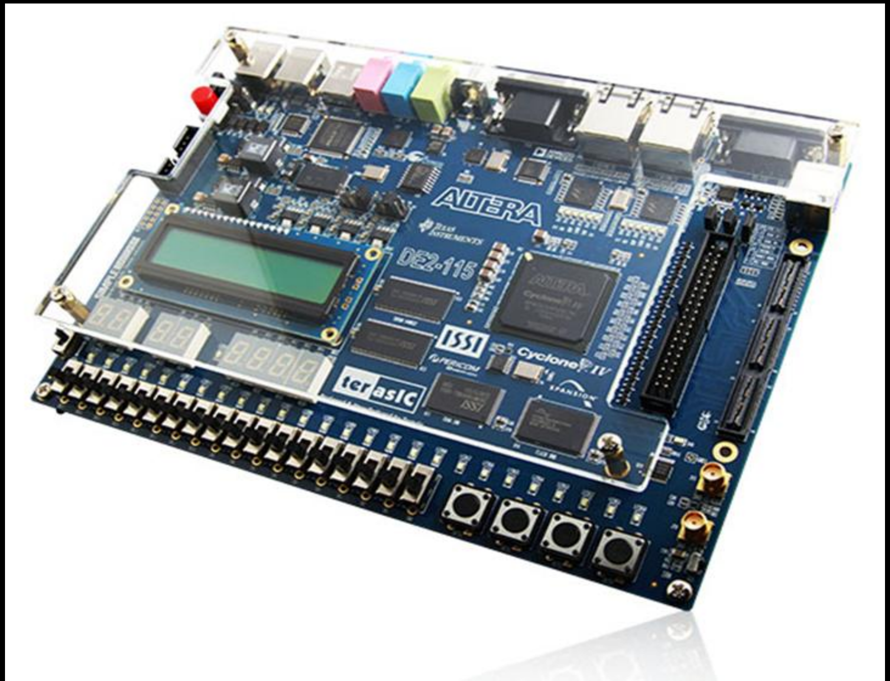
FPGA_50_2_4

FPGA Platform: DE2-115

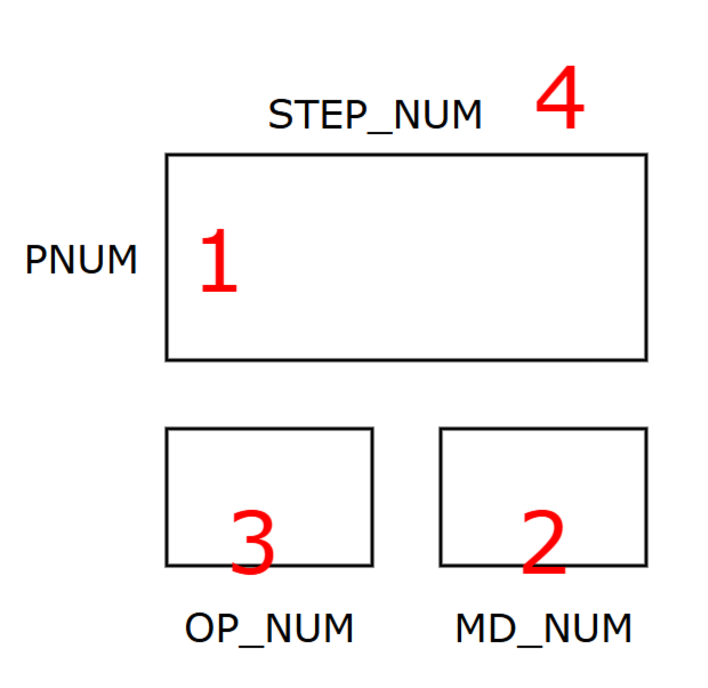
Resource utility: 22%

Clock frequency: 50 MHz

CoreMark/MHz: 5.90



+



FPGA_50_1_4

FPGA Platform: DE2-115

Resource utility: 9%

Clock frequency: 50 MHz

CoreMark/MHz: 3.47